



Review

Potential of resveratrol in mitigating metabolic disturbances induced by ethanol



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ARTICLE INFO

Keywords:

Resveratrol

Alcohol

Metabolism

Liver damage

ABSTRACT

Alcohol abuse is associated with numerous health problems, including metabolic disturbances and liver damage. Therefore, different compounds are continuously being tested to evaluate their potential effectiveness in reducing these harmful changes. Animal studies clearly show that resveratrol is capable of ameliorating some consequences of ethanol ingestion. Resveratrol is a naturally occurring diphenolic compound having pleiotropic, health-promoting properties. Its beneficial action have been also demonstrated in animal models with ethanol-induced metabolic disturbances and liver injury. In ethanol treated animals, resveratrol effectively reduced liver lipid accumulation. Moreover, this compound diminished necrosis of hepatocytes, and also reduced liver fibrosis. The hepatoprotective action of resveratrol is largely associated with its ant-oxidant and anti-inflammatory properties, and also covers changes in activities of some enzymes. It is known that this compound upregulates the adiponectin-SIRT1-AMPK signaling pathway in the liver. Resveratrol was also found to positively affect blood lipids in animals exposed to ethanol. Moreover, administration of resveratrol to animals with ethanol-induced hypoinsulinemia and insulin resistance was shown to alleviate these disturbances.

These outcomes clearly indicate that resveratrol holds great potential to reduce some consequences of ethanol ingestion. However, human studies are required to fully assess its therapeutic value.

1. Introduction

Alcohol abuse is a serious global problem associated with risk of many diseases, and also with increased morbidity and mortality. The number of alcohol-attributable diseases is very big and includes, among others, cancer, liver cirrhosis, pancreatitis, cardiovascular diseases, metabolic disorders, diabetes, infections, stroke, mental disorders and more [1]. Therefore, new synthetic drugs, and also naturally occurring compounds are continually being tested to prevent and better treat consequences of alcohol abuse [2]. Recently, there is a growing body of evidence that resveratrol, one of the naturally occurring agents, has a great potential in mitigating metabolic consequences of ethanol ingestion.

Resveratrol (3,5,4'-trihydroxystilbene, Fig. 1) is a polyphenol found in different plant species, among others in grapes, berries and peanuts. The richest natural source of resveratrol is *Polygonum cuspidatum*, a plant that has been used in oriental folk medicine. This compound is also present in red wine and has been proposed to be responsible for the "French paradox", i.e. low mortality due to cardiovascular disease as a result of moderate consumption of red wine [3,4]. Resveratrol belongs to the biologically active agents, and its ingestion is associated with

several benefits in the organism. Resveratrol rises great interest, in particular due to its health-promoting properties. Numerous favorable effects of this compound have been shown not only in animal studies, but also in humans. Its action is pleiotropic and involves, among others, anti-cancer [5], cardio-protective [6], anti-inflammatory [7], anti-diabetic [8,9], neuro-protective [10], anti-obesity [11] and anti-oxidative [12] effects. Moreover, experiments with mice on a high-calorie diet revealed the capacity of resveratrol to increase the survival and motor function. These effects were also accompanied by changes in expression of some genes and by reduction in fat depots [13,14].

At the cellular level, resveratrol is known to positively affect energy expenditure, mitochondrial biogenesis and cell survival. Its molecular targets involve many intracellular signaling molecules and regulatory proteins, including AMP-activated protein kinase (AMPK) and silent information regulator 1 (SIRT1) [15].

In the light of numerous benefits of resveratrol ingestion that have been documented under various pathological conditions, data from the literature showing the capability of this compound in ameliorating consequences of ethanol exposure seem to be very intriguing and create new perspectives for its use. This review summarizes results of studies addressing to the action of resveratrol in rodents exposed to ethanol

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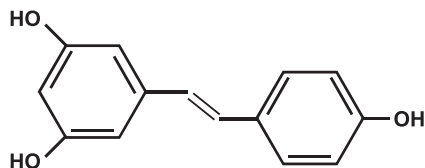


Fig. 1. Chemical structure of *trans*-resveratrol.

(Fig. 2).

2. Hepatoprotective action of resveratrol

In mammals, liver is the main organ in which ethanol undergoes degradation and thereby hepatocytes are subjected to damage by ethanol, and also by products of its metabolism. Oxidative metabolism of ethanol to acetate involves two steps. The first one is conversion to acetaldehyde. This reaction is catalyzed largely by alcohol dehydrogenase (EC 1.1.1.1), and also by catalase and microsomal enzymes. The second step is conversion of acetaldehyde to acetate in the presence of aldehyde dehydrogenase (EC 1.2.1.3). The mechanism underlying the hepatotoxic action of ethanol is complex and is attributed to many factors, including increased formation of NADH, generation of reactive oxygen species, induction of inflammation and more [16,17]. The degree of liver injury strongly depends on doses of ethanol and term of its ingestion.

2.1. Effects of resveratrol on liver lipids

Increased accumulation of fat in hepatocytes is the earliest response

Table 1
Effects of resveratrol on blood and liver lipids in ethanol-treated animals.

Parameter	Ethanol	Resveratrol	Animal	Ref
<i>Blood</i>				
Triglycerides	↑ 10%, drinking water, 2 w	↓ 10 mg, ig., 2 w	Rat	[23]
	2.4 g/kg bw, liquid diet, 4 w	100 mg, 22 w	Rat	[24]
	200 mg, oral, 4 w	30 or 100 mg	Mouse	[20]
		100 mg, oral, 4 w	Mouse	[19]
	liquid diet	200 or 400 mg, 2 w	Mouse	[18]
Free fatty acids	↑ 10%, drinking water, 2 w	↓ 10 mg, ig., 2 w	Rat	[23]
Total cholesterol	↑ 2.4 g/kg bw, liquid diet	↓ 100 mg, 22 w	Rat	[24]
LDL-cholesterol	liquid diet	30 or 100 mg	Mouse	[20]
<i>Liver lipids</i>	↑ 10%, drinking water, 2w	↓ 10 mg, ig., 2 w	Rat	[23]
	liquid diet, 4w	200 or 400 mg, 2 w	Mouse	[18]
	3 g/kg bw, ip, 6w	5 g/kg diet	Rat	[21]
	liquid diet	30 or 100 mg	Mouse	[20]
	200 mg, oral, 4w	100 mg, oral 4 w	Mouse	[19]
	6.5 g/kg bw, ig 4 w	250 mg, ig, 4 w	Rat	[22]

mg – mg/kg body weight, bw – body weight, w – weeks, ip – intraperitoneal, ig – intragastric.
LDL – low density lipoproteins.
↑ - increase, ↓ - decrease.

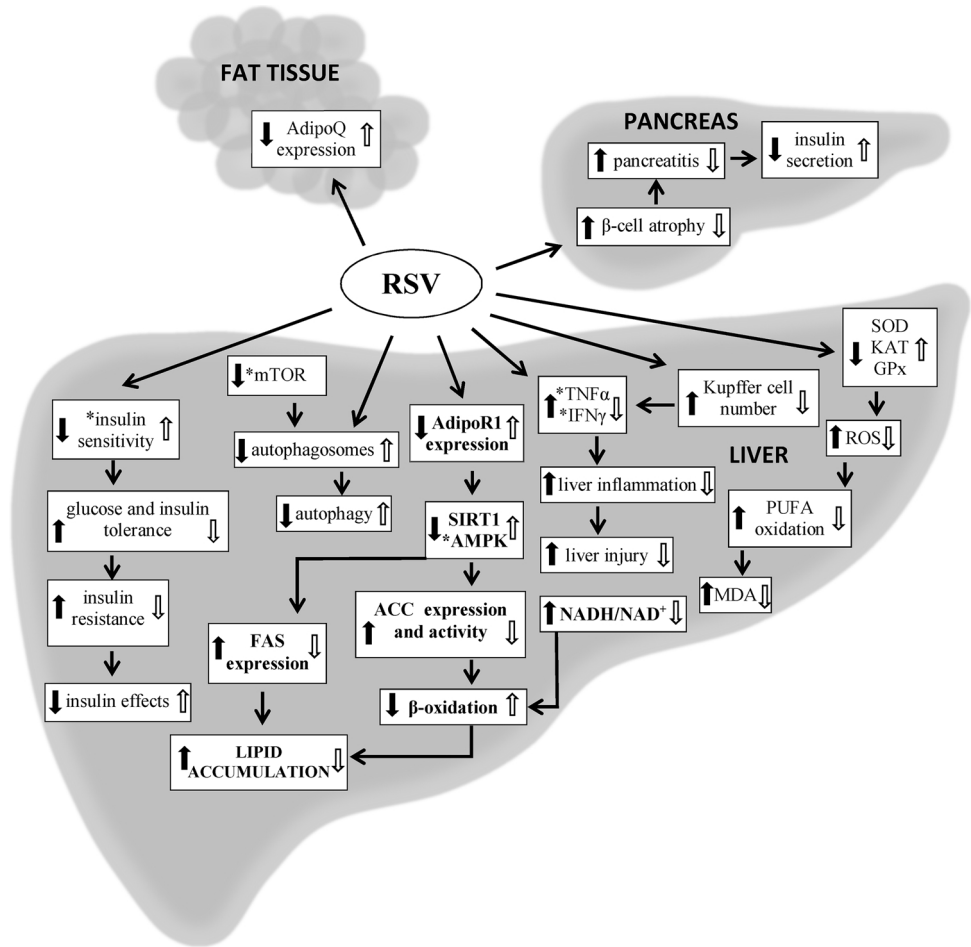


Fig. 2. Proposed mechanisms of resveratrol action in ethanol-treated animals. Black arrows - effects of ethanol; open arrows - effects of resveratrol. * - chronic ethanol ingestion.

to alcohol abuse [17]. It is known that exposure to ethanol stimulates the uptake of free fatty acids and reduces the rate of β -oxidation in hepatocytes. The latter effect is largely due to the exaggerated formation of intracellular NADH, since many enzymes catalyzing β -oxidation is inhibited as a result of increased NADH/NAD⁺ ratio [17]. Importantly, ethanol-induced liver lipid accumulation is known to be reversible and may be prevented by some naturally occurring compounds [2]. In the last years, rodent studies have shown that resveratrol is capable of reducing ethanol-induced fatty liver. It was demonstrated that ethanol increased liver lipid accumulation in mice [18–20] and rats [21–23], however, resveratrol treatment effectively decreased liver lipid content in these animals (Table 1). Resveratrol-induced decrease in liver lipid accumulation, shown in ethanol-treated animals, is achieved due to multiple effects. One is the influence on enzymes related with lipid metabolism. It was demonstrated that ethanol increased expression of liver fatty acid synthase (FAS) and acetyl-CoA carboxylase (ACC), whereas resveratrol down-regulated expression of these enzymes [18]. Moreover, resveratrol administered to rats fed an ethanol containing diet, markedly diminished liver NADH/NAD⁺ ratio in these animals [24]. This is an important finding given that decrease in NADH/NAD⁺ ratio promotes β -oxidation and thereby reduces liver lipid accumulation.

Ajmo et al. [18] have proposed that the protective effects of resveratrol against ethanol-induced liver damage are largely mediated by adiponectin, an adipokine having multiple regulatory functions. The insufficient secretion and/or action of adiponectin is associated, among others, with insulin resistance and liver dysfunction. Chronic ethanol intake reduces expression of adiponectin receptors in the liver, down-regulates adiponectin expression in fat tissue and is accompanied by hypoadiponectinemia. In consequence, the impaired action of this adipokine contributes to reduced fatty acid oxidation and increased lipid accumulation in the liver. Role of adiponectin in alcoholic fatty liver disease is well established in animal models [25]. Consistent with this notion, it was shown that in ethanol-treated mice, resveratrol improved adiponectin action in the liver [18]. It is thought that the positive effects of this adipokine on liver are strongly linked with SIRT1 and AMPK. Adiponectin confers protection against alcoholic fatty liver via modulation of complex hepatic processes controlled by the adiponectin-SIRT1-AMPK pathway [26]. Therefore, impairment in this pathway is associated with alcoholic fatty liver disease [16,25,27]. SIRT1 is NAD⁺-dependent histone deacetylase involved in the regulation of many processes, such as mitochondrial biogenesis, inflammation, intracellular metabolism, stress resistance and apoptosis. AMPK is an intracellular energy sensor. Activation of this enzyme inhibits anabolic pathways and simultaneously promotes catabolic processes to increase intracellular ATP pool. Once activated, AMPK (pAMPK) phosphorylates downstream targets, including acetyl-CoA carboxylase (ACC). Phosphorylation of ACC by pAMPK is at an inhibitory site and thereby is linked to decreased conversion of acetyl-CoA to malonyl-CoA, finally promoting β -oxidation of long-chain fatty acids [28] and leads to reduced lipid accumulation in tissues. However, under some pathological conditions, the action of AMPK and related enzymes may be disturbed. Moreover, effects of ethanol ingestion on AMPK in the liver effects may be different. Ethanol may exert not significant influence on pAMPK [23], but it also may increase [29,30] or reduce [18,27,31] phosphorylation and activity [32] of AMPK. These data are, however, not contradictory, since effects of ethanol on AMPK largely depend on time of its ingestion. Chronic ingestion down-regulates AMPK in hepatocytes. Other effects are observed after short-term treatment. It was demonstrated that in mice exposed to ethanol for 22 weeks, hepatic pAMPK α , AMPK α , pAMPK β and AMPK β were reduced, however, resveratrol normalized these changes [18]. Recently, Luo et al. [24] have reported that in response to long-term ethanol administration to rats, liver SIRT1 expression was lower compared with control subjects. This negative change was effectively prevented by resveratrol. This is a relevant finding given the role of SIRT1 in the

mechanism of ethanol-induced liver damage. It is known that ethanol down-regulates expression/activity of SIRT1 in hepatocytes, and this contributes to reduced β -oxidation of fatty acids and to increased liver lipid accumulation [26].

Decrease in liver lipid accumulation by resveratrol treatment is undoubtedly very relevant, since in the case of prolonged ethanol ingestion, fatty liver often leads to liver inflammation and to more severe liver injury *i.e.* fibrosis and necrosis. It is also known that, ethanol-induced disturbances in liver lipid metabolism are often associated with dyslipidemia. In humans, effects of alcohol on blood lipids differ depending on various conditions, including genetic differences, pattern of alcohol use, age, sex and more [33]. Rodent studies show that parameters of lipid metabolism in the blood are negatively affected by ethanol treatment, however, resveratrol is capable of ameliorating these adverse changes. Luo et al. [24] have shown that chronic ethanol feeding of rats increased blood levels of total cholesterol, however, this effect was prevented by resveratrol. It was also demonstrated that in ethanol-treated mice and rats, blood levels of low density lipoprotein cholesterol were pathologically elevated, however, resveratrol treatment reduced LDL-cholesterol [20,24]. Resveratrol administration to rats and mice subjected to ethanol was also found to reduce blood triglyceride levels [18–20,23,24]. It was also reported that ethanol increased blood free fatty acid levels, whereas in response to resveratrol ingestion, this effect was attenuated [23] (Table 1).

2.2. Effects of resveratrol on liver inflammation

Liver inflammation is usually associated with chronic alcohol consumption [34]. Therefore, anti-inflammatory compounds could be effective in reducing long-term consequences of alcohol abuse. In the case of resveratrol, its anti-inflammatory properties are well documented in different animal models [7] and have also been shown in rats and mice exposed to ethanol [21,35,36]. It is thought that inflammation and innate immunity have an important role in the pathogenesis of alcoholic liver disease [17]. Therefore, anti-inflammatory properties of resveratrol undoubtedly contribute to its protective action against ethanol-induced liver injury. This action is largely associated with inflammatory cytokines, in particular TNF- α and INF- γ . There are produced predominantly by Kupffer cells and contribute to ethanol-induced liver injury. It is also known that acetaldehyde induces translocation of lipopolysaccharide and both these compounds promote production of proinflammatory cytokines by Kupffer cells. Results of long-term studies show that in mice exposed to ethanol, resveratrol diminished the number of Kupffer cells [35], and also significantly decreased blood levels of TNF α and INF γ [36].

2.3. Effects of resveratrol on liver oxidative damage

Apart from anti-inflammatory properties of resveratrol, its anti-oxidative effects have been documented to play a role in the protection against alcoholic liver injury. Oxidative metabolism of ethanol leads to the generation of reactive oxygen species (ROS), *i.e.* hydrogen peroxide and superoxide radical. It is well established that ROS have a major role in the mechanism underlying the hepatotoxic action of ethanol [17]. Despite the presence of the anti-oxidant defense system, the increased formation of ROS, particularly in the case of chronic ethanol intake, leads to damage of cellular lipids, proteins and nucleic acids. The enzymatic anti-oxidant defense system includes superoxide dismutase (SOD, EC 1.15.1.1), which catalyzes the dismutation of the superoxide radical, catalase (CAT, EC 1.11.1.6), which participates in the decomposition of hydrogen peroxide to water and to oxygen, and also glutathione peroxidase (GPx, EC 1.11.1.9), catalyzing reduction of lipid hydroperoxides and free hydrogen peroxide.

It is noteworthy that DNA damage, *eg.* by toxins and ROS, activates the DNA damage response and repair signaling cascade. Stimulation of this cascade triggers, in turn, immune response. It is recently well

established that relationship between the DNA damage and immune response is bidirectional [37]. This connection suggests that protective action on DNA may be associated with reduced immune response and inflammation and *vice versa*. Various anti-oxidants, including resveratrol, may effectively protect liver cells against ethanol-induced injury. In the case of resveratrol, its anti-inflammatory [21,35,36] and anti-oxidant [18,21,38] properties have been shown to play an important role in the protection against ethanol action. Given the bidirectional relationship between DNA damage and immune response [37], anti-oxidant action of resveratrol, resulting in reduced DNA damage, may also attenuate inflammatory response. Moreover, anti-inflammatory action of resveratrol may lead to reduced DNA damage.

The anti-oxidative capacity of resveratrol is well established in different experimental models [12]. Importantly, its anti-oxidant effectiveness has been also shown in rats and mice exposed to ethanol. These effects are associated with changes in the activities of enzymes of the anti-oxidant defense system. Studies addressing the action of resveratrol demonstrated that ethanol treatment reduced activities of SOD, CAT and GPx in the rat liver, whereas resveratrol administered to these animals significantly increased their activities [21]. In parallel, in ethanol-treated subjects, resveratrol was also shown to augment blood activities of SOD, CAT and GPx [36] (Table 2).

Apart from favorable influence on enzyme activities, more direct evidence for the role of the anti-oxidative capacity of resveratrol in the mechanism underling its hepatoprotective action has also been provided. It was found that production of ROS by liver mitochondria isolated from rats exposed to ethanol was increased, however, this harmful effect was markedly attenuated in the animals co-treated with resveratrol [22]. Moreover, resveratrol was shown to positively affect concentrations of malondialdehyde (MDA). This compound is a product of lipid peroxidation and, therefore is an important marker of oxidative stress. Ethanol ingestion is associated with increased tissue MDA levels [17]. Importantly, results of several studies have shown that contents of MDA in the liver [18,21,38] and blood [19,36] of ethanol-treated animals were significantly decreased in response to resveratrol treatment (Table 2). Additionally, Raal et al. [35] revealed the capacity of

resveratrol to reduce the oxidation of polyunsaturated fatty acids in mice exposed to ethanol. These properties of resveratrol are of great relevance given that lipid peroxidation, with a concomitant formation of cytotoxic MDA, is thought to be one of the most important processes involved in ethanol-induced liver damage [17].

Anti-oxidative properties of resveratrol may have positive implications on liver lipid accumulation *via* reduced damage of nucleic acids and changes in activities of some enzymes.

Oxidative metabolism of ethanol is associated with damage of nucleic acids. Consequences of this differ depending on degree of damage. In the case of a large oxidative stress, cell apoptosis may be induced. Moreover, chronic alcohol ingestion leads to DNA hypomethylation and damage of mtDNA. Chronic ethanol ingestion may also affect a small noncoding micro RNA. This effect may have metabolic consequences, since reduces expression of SIRT1, which in turn increases expression of lipogenic enzymes in hepatocytes, *i.e.* acetyl-CoA carboxylase, fatty acid synthase and stearoyl-coenzyme A desaturase 1, and decreases expression of genes involved in fatty acid oxidation [39]. Therefore, it can be supposed that beneficial effects of resveratrol on liver lipid accumulation are largely due to reduced oxidative damage of nucleic acids in hepatocytes.

2.4. Effects of resveratrol on autophagy and fibrosis

Another effect whereby resveratrol exerts its hepatoprotective action against ethanol-induced injury is regulation of autophagy [20]. Autophagy is a process that involves intracellular degradation of damaged organelles and proteins to promote cell survival. Autophagy regulates lipid droplets in hepatocytes and is also an important process contributing to attenuation of lipid accumulation in these cells. mTOR (the mammalian target of rapamycin) is the main regulator of autophagy. Intracellular stress with a concomitant inhibition of mTOR activates autophagy. Long-term ethanol ingestion reduces autophagy, but short-term exposure to ethanol up-regulates this process [17]. Tang et al. [20] have reported that in mice fed a diet with ethanol, resveratrol increased the number of autophagosomes and promoted the formation of

Table 2

Effects of resveratrol on activities of enzymes related with liver damage and parameters of antioxidant defense in ethanol-treated animals.

Parameter		Ethanol		Resveratrol	Animal	Ref
Blood						
Gamma glutamyltransferase	↑	10%, drinking water, 2 w	↓	10 mg, ig, 2 w	Rat	[23]
		3 g/kg bw, ip, 6 w		5 g/kg diet	Rat	[21]
Aspartate aminotransferase	↑	3 g/kg bw, ip, 6 w	↓	5 g/kg diet	Rat	[21]
		1.6 g/kg bw, oral, 12 w		5 mg, 12 w	Mouse	[35]
		liquid diet		30 or 100 mg	Mouse	[20]
		6.5 g/kg bw, ig, 4 w		250 mg, ig, 4 w	Rat	[22]
Alanine aminotransferase	↑	3 g/kg bw, ip, 6 w	↓	5 g/kg diet	Rat	[21]
		1.6 g/kg bw, oral, 12 w		5 mg, 12 w	Mouse	[35]
		liquid diet		30 or 100 mg	Mouse	[20]
		6.5 g/kg bw, ig, 4 w		250 mg, ig, 4 w	Rat	[22]
		liquid diet, 4 w		200 or 400 mg, 2 w	Mouse	[18]
Alkaline phosphatase	↑	3 g/kg bw, ip, 6 w	↓	5 g/kg diet	Rat	[21]
Liver						
Superoxide dismutase	↓	3 g/kg bw, ip, 6 w	↑	5 g/kg diet	Rat	[21]
Catalase	↓	3 g/kg bw, ip, 6 w	↑	5 g/kg diet	Rat	[21]
Glutathione peroxidase	↓	3 g/kg bw, ip, 6 w	↑	5 g/kg diet	Rat	[21]
Malondialdehyde	↑	3 g/kg bw, ip, 6 w	↓	5 g/kg diet	Rat	[21]
		3 g/kg bw, ip, 6 w		5 g/kg diet, 6 w	Rat Mouse	[36] [18]
		liquid diet, 4 w		200 or 400 mg, 2 w		
Blood						
Superoxide dismutase	↓	1.6 g/kg bw, oral, 12 w	↑	5 mg	Mouse	[35]
Catalase	↓	1.6 g/kg bw, oral, 12 w	↑	5 mg	Mouse	[35]
Glutathione peroxidase	↓	1.6 g/kg bw, oral, 12 w	↑	5 mg	Mouse	[35]
TBARS Malondialdehyde	↑	1.6 g/kg bw, oral, 12 w	↓	5 mg	Mouse	[35]
		200 mg, oral, 4 w		100 mg, oral, 4 w	Mouse	[19]

mg – mg/kg body weight, bw – body weight, w – weeks, ip – intraperitoneal, ig – intragastric.

TBARS – thiobarbituric acid reactive substances.

↑ – increase, ↓ – decrease.

autophagy. Importantly, resveratrol-induced activation of autophagy was accompanied by reduced liver steatosis in ethanol-fed subjects.

It is known that resveratrol is capable of ameliorating not only early symptoms of ethanol ingestion, such as liver lipid accumulation. This compound, given to ethanol-treated animals, was also demonstrated to alleviate more profound changes. Kasdallah-Grissa et al. [21] have shown that in rats treated with ethanol, resveratrol markedly diminished necrosis of hepatocytes, and also reduced liver fibrosis. Alcohol-induced hepatic fibrosis, mainly promoted by acetaldehyde, is associated with deposition of fibrillar collagens [17].

2.5. Effects of resveratrol on blood markers of liver damage

Additional evidence for the hepatoprotective action of resveratrol in ethanol treated animals is the influence of this compound on blood markers of liver injury, i.e. gamma glutamyltransferase, aspartate aminotransferase, alanine aminotransferase, and also alkaline phosphatase. Gamma glutamyltransferase (GGT, EC 2.3.2.2) catalyzes the transfer of gamma-glutamyl functional groups from molecules such as glutathione to an acceptor that may be an amino acid, a peptide or water. GGT is one of the traditional markers of alcohol abuse. Blood activities of GGT increase in response to ethanol consumption [40]. Aspartate aminotransferase (AST, EC 2.6.1.1) and alanine aminotransferase (ALT, EC 2.6.1.2) are involved in metabolism of amino acids and are sensitive indicators of ethanol-induced liver cell injury. Alkaline phosphatase (ALP, EC 3.1.3.1) catalyzes dephosphorylation of different compounds, is less specific marker of liver injury. Several studies have reported that ethanol treatment of rats and mice increased blood activities of GGT [21,23], ALT [18,20–22,36,41], AST [20–22,36,41] and ALP [21] in these animals. However, resveratrol significantly reduced blood activities of these enzymes in ethanol-ingesting subjects (Table 2).

3. Effects of resveratrol on blood insulin levels and insulin action

Ethanol ingestion may be associated with chronic pancreatitis, atrophy of pancreatic β -cells and reduced secretion of insulin [23,42,41]. Recent data show that resveratrol markedly alleviates ethanol-induced decrease in blood insulin levels, and also improves insulin/glucagon ratio [23]. This indicates the protective effects of resveratrol on the endocrine pancreas in ethanol-ingesting animals. Advantageous effects of resveratrol on the endocrine pancreas have been previously shown in different experimental models, namely in rodents on a high-fat diet [44], carboxymethyllysine-induced pancreas dysfunction, streptozotocin-nicotinamide induced diabetes [45], and also in db/db mice [46].

Apart from affecting blood insulin levels, ethanol may also influence insulin action on target tissues. It is thought that moderate alcohol consumption may improve insulin sensitivity [47]. However, chronic alcohol abuse is associated not only with dysfunction of pancreatic β -cells, but also with developing insulin resistance [43], both being the hallmark of type 2 diabetes. It was revealed that long-term (22 weeks) ingestion of ethanol by rats resulted in reduced insulin sensitivity compared with control animals. Importantly, insulin action in these animals was markedly improved in response to resveratrol treatment. This effect was manifested by increased glucose and insulin tolerance in resveratrol-treated rats. Moreover, resveratrol increased expression of some proteins of the insulin signaling pathway in the liver of ethanol-treated rats. Finally, metabolic functioning of the liver was improved, since ethanol-induced decrease in liver glycogen content was alleviated by resveratrol [24].

Given the pivotal role of insulin in the regulation of numerous metabolic processes, beneficial impact of resveratrol on both blood insulin levels [23] and insulin action [24], certainly largely contributes to the normalization of metabolic disturbances in animals exposed to ethanol. This indicates that resveratrol mitigates metabolic

disturbances *via* direct action on some tissues, but also *via* improvement in insulin action.

4. Additional remarks

Exposure of experimental animals to ethanol is usually accompanied by decrease in body weight gains. The majority of available data shows that resveratrol treatment ameliorates this negative effect [21,24,36,38]. However, this is not a prerequisite for its action, since this compound was also demonstrated to induce numerous beneficial changes without significant influence on body weight gains [23,41].

In the context of the potential therapeutic use of resveratrol, it should be mentioned that this compound is well tolerated in humans and animals [48]. Amounts of resveratrol used in animal studies to effectively alleviate metabolic consequences of ethanol ingestion are, however, in relatively broad range (Tables 1 and 2). The lowest effective dose of resveratrol was 5 mg/kg body weight in mice [36] and 10 mg in the rat [23].

The available data show that resveratrol exerts beneficial effects when is administered for two weeks [23] or much longer (22 weeks) [24]. This indicates its efficacy in both short-term and long-term period. This is an important finding given that disturbances induced by alcohol abuse differ depending on term of ingestion [17]. In studies addressing the impact of resveratrol in ethanol treated rats or mice, the tested compound was demonstrated to be effective when was administered to animals once a day *via* oral gavage [19,22,23,36] or incorporated in a diet [18,38,40].

5. Conclusions and future perspectives

Results of studies summarized in this paper clearly show that resveratrol holds great potential to alleviate metabolic disturbances associated with ethanol ingestion. In particular, this compound was demonstrated to exert hepatoprotective action in animals with ethanol-induced liver injury. This action covers changes in the activities of some enzymes and is strongly linked with anti-oxidant and anti-inflammatory properties of resveratrol. Moreover, resveratrol positively affects blood insulin levels and insulin action. These data indicate that this compound could be helpful in ameliorating some consequences of alcohol abuse. Given that resveratrol has pleiotropic effects, it is difficult to show the exact mechanism underlying its action. This is, however, a positive feature, since different properties of this compound certainly complement each other providing much better protection (Fig. 2).

Although beneficial action of resveratrol have been relatively well established in animal models, human studies are necessary to evaluate its therapeutic value. Moreover, results of studies show that the effective doses of resveratrol in animals are very diverse, and the mechanism of its action may differ depending on doses. Treatment with lower doses may activate SIRT1, whereas higher amounts of resveratrol may activate AMPK in a SIRT1-independent manner [49]. Therefore, it would be advisable to determine the lowest effective doses in animals in the context of different mechanism of action.

Moreover, other aspects of resveratrol action addressing the consequences of alcohol abuse [1] remain to be elucidated. In the other experimental models, resveratrol was demonstrated to exert, among others, neuro-protective [10] and cardio-protective [6] effects. This compound has also anti-cancer properties [50] and may reduce pancreatitis [51]. However, potential of resveratrol in alleviating the alcohol-attributable diseases is still unknown. Clarification of all these issues may indicate a wider possibilities of resveratrol use in ameliorating consequences of alcohol abuse.

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